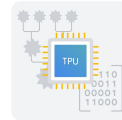


AI Infrastructure: Cloud TPUs

Congratulations on completing the course. This course summary is your review guide. Print it for a handy reference.

TPU stands for Tensor Processing Unit. These are custom-developed, application-specific integrated circuits (ASICs) built by Google specifically to accelerate the intensive computations found in machine learning. Here's when they're your best bet:



Training massive deep learning models

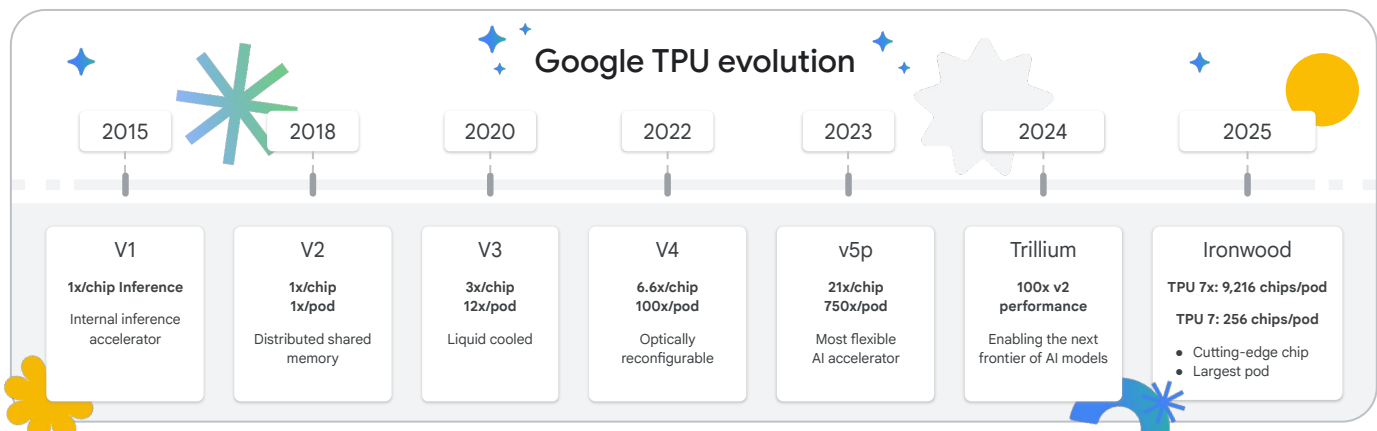


Models relying on embeddings



Scientific and healthcare AI

Google TPU evolution



GPU and TPU interoperability:

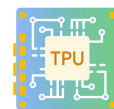
- To achieve seamless GPU and TPU switching, use a dual-container approach within a single pod.
- If the appropriate accelerator (GPU or TPU) is present, that container's vLLM server starts; otherwise, it sleeps. This ensures only the correct vLLM server is active based on the underlying hardware.

Dynamic Workload Scheduler (DWS)

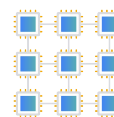
- Flex mode allows users to request hardware for specified periods (from 1 minute to 7 days).
- Calendar mode enables users to create future reservations for hardware they know they will need in advance.

Cloud TPU consumption options

- Long-term reservations allow you to request and reserve TPU resources in advance for an extended period.
- On-demand allows you to request TPU resources to be used as soon as possible, and you can keep them for as long as you want.
- Spot allows you to request TPU resources that could be preempted (shut down) by Google Cloud at any time if capacity is needed elsewhere.



A TPU pod is a collection of TPUs that are physically grouped together and connected by a specialized, high-speed network.



A slice is a subset of chips within a single TPU pod, all connected by incredibly fast Inter-Chip Interconnects (ICI).



TPU cube
A 4x4x4 topology of interconnected TPU chips.

Best practices for model development

Principle 1: Layout for efficiency

Principle 2: Fixed shapes for predictable performance

Principle 3: Avoiding unnecessary padding

Additional Resources

1. [AI Hypercomputer documentation](#)
2. [Ironwood: The first Google TPU for the age of inference](#)
3. [About TPUs in GKE](#)

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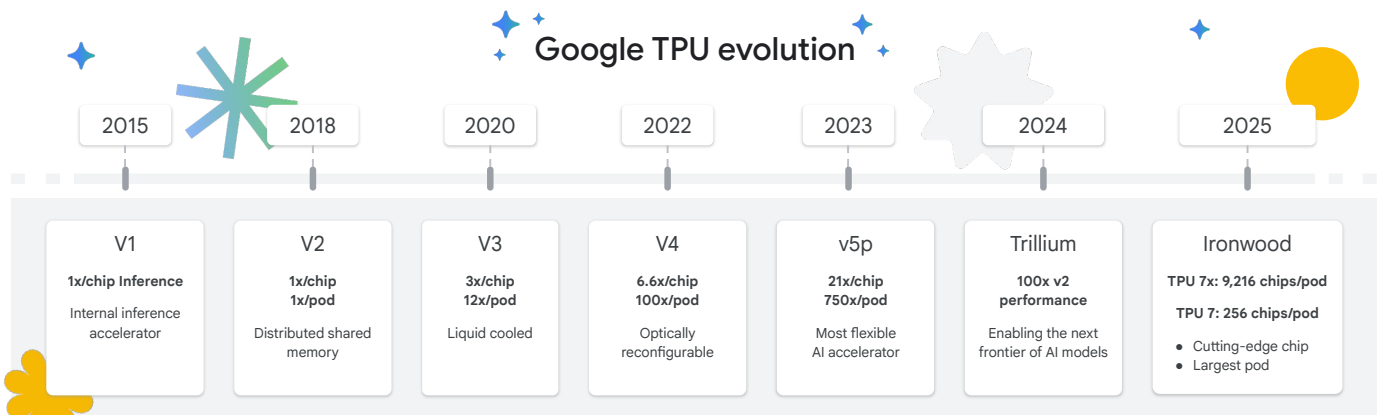


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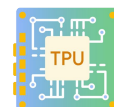
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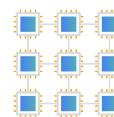
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